



COS20028 – BIG DATA ARCHITECTURE AND APPLICATION LAB WEEK 02

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Week 2



Running a
MapReduce
Job

Week 02 Activity

- You'll compile Java files, create a JAR, and run MapReduce jobs in this tutorial.

Files and Directories Used in this Exercise

Source directory: `~/workspace/wordcount/src/stubs`

Files:

`WordCount.java`: A simple MapReduce driver class.

`WordMapper.java`: A mapper class for the job.

`SumReducer.java`: A reducer class for the job.

`wc.jar`: The compiled, assembled WordCount program

A terminal window with a dark title bar showing 'root@192-168-1-109:~'. The menu bar includes 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The terminal content shows a user switching to root with 'sudo su -' and then typing 'shutdown -r now' with a cursor at the end.

```
root@192-168-1-109:~  
File Edit View Search Terminal Help  
[training@192-168-1-109 wordcounts]$ sudo su -  
[root@192-168-1-109 ~]# shutdown -r now
```

Reboot Your VM

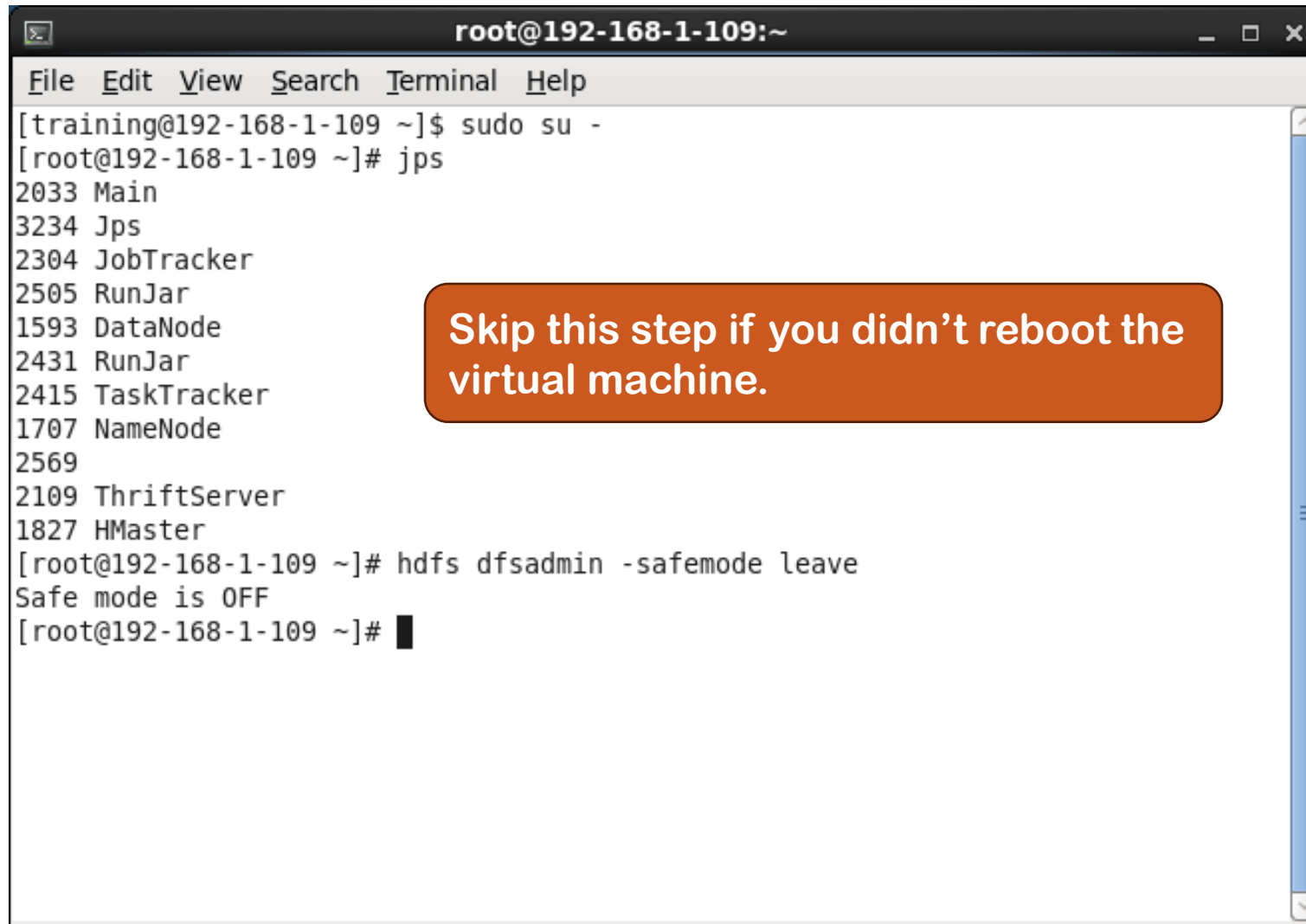
- `sudo su -`
- `shutdown -r now`



```
root@192-168-1-109:~  
File Edit View Search Terminal Help  
[training@192-168-1-109 ~]$ sudo su -  
[root@192-168-1-109 ~]# jps  
2033 Main  
3234 Jps  
2304 JobTracker  
2505 RunJar  
1593 DataNode  
2431 RunJar  
2415 TaskTracker  
1707 NameNode  
2569  
2109 ThriftServer  
1827 HMaster  
[root@192-168-1-109 ~]#
```

Check Your Running Demands

- `sudo su -`
- `jps`
- **Note:** You should see **NameNode** and **DataNode** running.



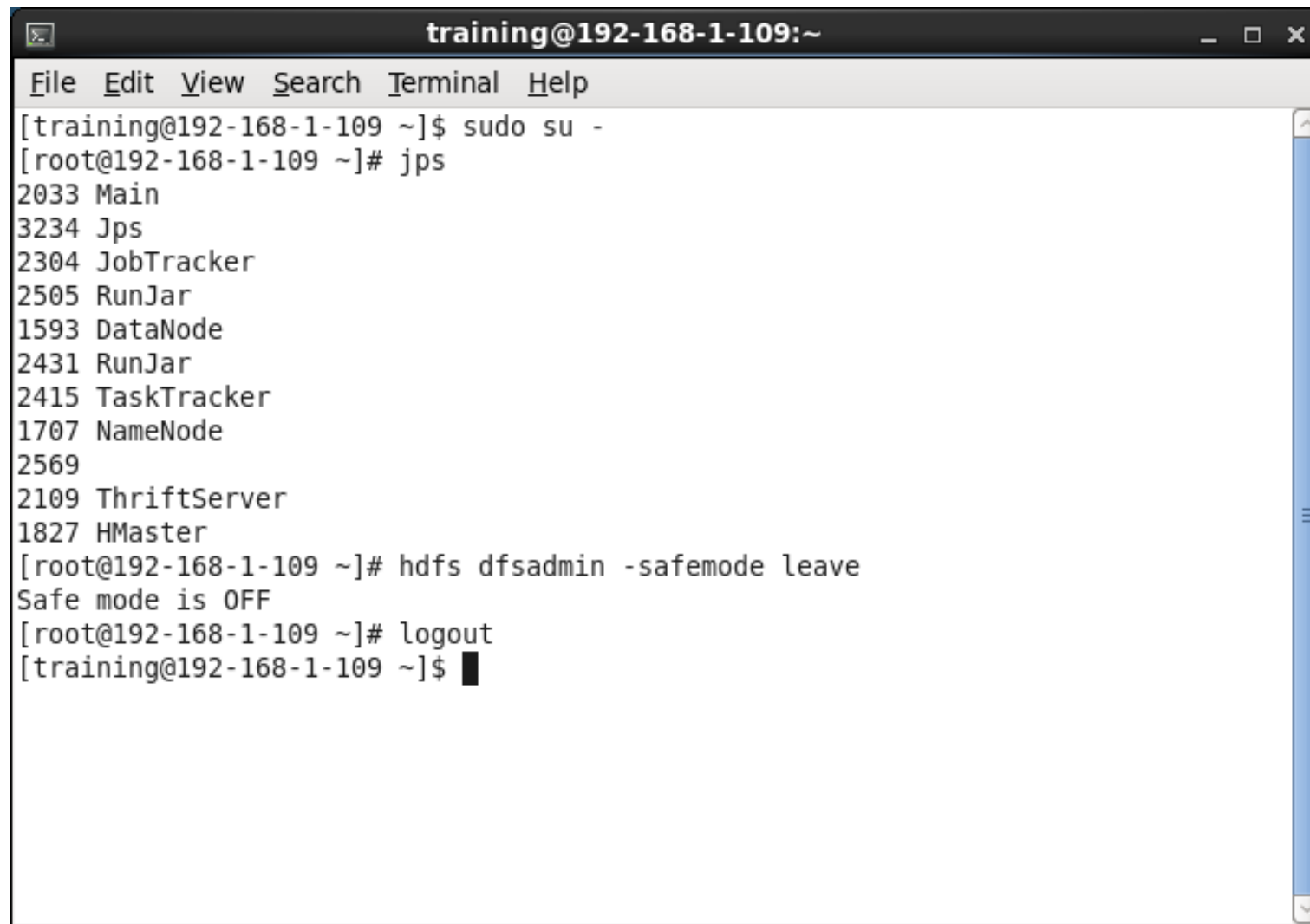
A terminal window titled 'root@192-168-1-109:~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

```
[training@192-168-1-109 ~]$ sudo su -  
[root@192-168-1-109 ~]# jps  
2033 Main  
3234 Jps  
2304 JobTracker  
2505 RunJar  
1593 DataNode  
2431 RunJar  
2415 TaskTracker  
1707 NameNode  
2569  
2109 ThriftServer  
1827 HMaster  
[root@192-168-1-109 ~]# hdfs dfsadmin -safemode leave  
Safe mode is OFF  
[root@192-168-1-109 ~]#
```

An orange callout box with white text is overlaid on the terminal output, stating: "Skip this step if you didn't reboot the virtual machine."

Exit from the Safemode

- `hdfs dfsadmin -safemode leave`



A terminal window titled "training@192-168-1-109:~" with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

```
[training@192-168-1-109 ~]$ sudo su -  
[root@192-168-1-109 ~]# jps  
2033 Main  
3234 Jps  
2304 JobTracker  
2505 RunJar  
1593 DataNode  
2431 RunJar  
2415 TaskTracker  
1707 NameNode  
2569  
2109 ThriftServer  
1827 HMaster  
[root@192-168-1-109 ~]# hdfs dfsadmin -safemode leave  
Safe mode is OFF  
[root@192-168-1-109 ~]# logout  
[training@192-168-1-109 ~]$
```

Exit root

- Ctrl + D

```
training@192-168-1-109:~  
File Edit View Search Terminal Help  
[training@192-168-1-109 ~]$ hdfs fsck /user/training  
Connecting to namenode via http://0.0.0.0:50070  
FSCK started by training (auth:SIMPLE) from /127.0.0.1 for path /user/training at Mon Aug 03 13:59:47 EDT 2020  
.....Status: HEALTHY  
Total size:      5343545 B  
Total dirs:      3  
Total files:      7  
Total blocks (validated):      7 (avg. block size 763363 B)  
Minimally replicated blocks:    7 (100.0 %)  
Over-replicated blocks:         0 (0.0 %)  
Under-replicated blocks:        0 (0.0 %)  
Mis-replicated blocks:          0 (0.0 %)  
Default replication factor:     1  
Average block replication:      1.0  
Corrupt blocks:                 0  
Missing replicas:               0 (0.0 %)  
Number of data-nodes:           1  
Number of racks:                1  
FSCK ended at Mon Aug 03 13:59:47 EDT 2020 in 6 milliseconds  
  
The filesystem under path '/user/training' is HEALTHY  
[training@192-168-1-109 ~]$
```

Check File System Status

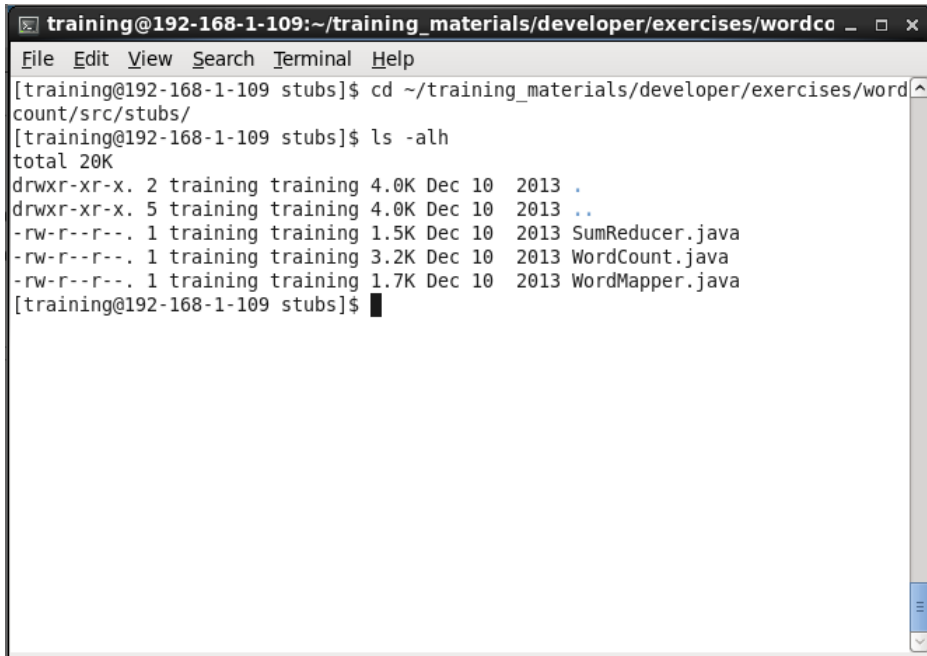
- `hdfs fsck /user/training`
- If your HDFS is not in a HEALTHY state, you'll encounter issues when running MapReduce.
- Solution: Remove all files on Hadoop and reupload.


```
training@192-168-1-109:~  
File Edit View Search Terminal Help  
.....Status: HEALTHY  
Total size: 5343545 B  
Total dirs: 3  
Total files: 7  
Total blocks (validated): 7 (avg. block size 763363 B)  
Minimally replicated blocks: 7 (100.0 %)  
Over-replicated blocks: 0 (0.0 %)  
Under-replicated blocks: 0 (0.0 %)  
Mis-replicated blocks: 0 (0.0 %)  
Default replication factor: 1  
Average block replication: 1.0  
Corrupt blocks: 0  
Missing replicas: 0 (0.0 %)  
Number of data-nodes: 1  
Number of racks: 1  
FSCK ended at Mon Aug 03 13:59:47 EDT 2020 in 6 milliseconds  
  
The filesystem under path '/user/training' is HEALTHY  
[training@192-168-1-109 ~]$ hdfs dfs -ls  
Found 2 items  
drwxr-xr-x - training supergroup 0 2020-08-03 13:30 shakespeare  
drwxr-xr-x - training supergroup 0 2020-08-03 13:24 wctest  
[training@192-168-1-109 ~]$
```

Check Target Folder

- `hdfs dfs -ls`

File Location

A terminal window titled 'training@192-168-1-109:~/training_materials/developer/exercises/wordco _' with standard window controls. The terminal shows a sequence of commands: 'cd ~/training_materials/developer/exercises/wordcount/src/stubs/' and 'ls -alh'. The output of 'ls -alh' lists directory entries with permissions, owner, group, size, date, and name. The files listed are '.', '..', 'SumReducer.java', 'WordCount.java', and 'WordMapper.java'.

```
training@192-168-1-109:~/training_materials/developer/exercises/wordco _
File Edit View Search Terminal Help
[training@192-168-1-109 stubs]$ cd ~/training_materials/developer/exercises/wordcount/src/stubs/
[training@192-168-1-109 stubs]$ ls -alh
total 20K
drwxr-xr-x. 2 training training 4.0K Dec 10 2013 .
drwxr-xr-x. 5 training training 4.0K Dec 10 2013 ..
-rw-r--r--. 1 training training 1.5K Dec 10 2013 SumReducer.java
-rw-r--r--. 1 training training 3.2K Dec 10 2013 WordCount.java
-rw-r--r--. 1 training training 1.7K Dec 10 2013 WordMapper.java
[training@192-168-1-109 stubs]$
```

- **cd**
~/training_materials/developer/exercises/wordcount/src/stubs
- **ls -alh**

File Explanation

- **WordCount.java**
 - A simple MapReduce driver class.
- **WordMapper.java**
 - A mapper class for the job.
- **SumReducer.java**
 - A reducer class for the job.
- **Examine these files if you wish, but do not change them.**

Examine the classpath configured in Hadoop

- hadoop classpath
- This shows lists the locations where the Hadoop core API classes are installed.

A terminal window titled 'training@192-168-1-109:~/training_materials/developer/exercises/wordco' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'hadoop classpath' and its output, which lists various Hadoop-related library paths separated by colons. The output is: /etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/.*:/usr/lib/hadoop-hdfs/.*:/usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/.*:/usr/lib/hadoop-yarn/lib/*:/usr/lib/hadoop-yarn/.*:/usr/lib/hadoop-0.20-mapreduce/.*:/usr/lib/hadoop-0.20-mapreduce/lib/*:/usr/lib/hadoop-0.20-mapreduce/.*. The prompt is [training@192-168-1-109 hints]\$ and there is a cursor at the end.

```
training@192-168-1-109:~/training_materials/developer/exercises/wordco
File Edit View Search Terminal Help
[training@192-168-1-109 hints]$ hadoop classpath
/etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/.*:/usr/lib/hadoop-hdfs
/.*:/usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/.*:/usr/lib/hadoop-yarn/li
b/*:/usr/lib/hadoop-yarn/.*:/usr/lib/hadoop-0.20-mapreduce/.*:/usr/lib/hadoop-
0.20-mapreduce/lib/*:/usr/lib/hadoop-0.20-mapreduce/.*
[training@192-168-1-109 hints]$
```


backquote

backquote

```
[training@192-168-1-109 src]$  
[training@192-168-1-109 src]$ javac -classpath `hadoop classpath` stubs/*.java  
[training@192-168-1-109 src]$
```

Compile the Java Classes

- `cd`
~/training_materials/d
eveloper/exercises/wo
rdcount/src
- `javac -classpath`
`hadoop classpath`
stubs/*.java



Collect Your Compiled Java Files into JAR File

- `jar cvf wc.jar stubs/*.class`

```
training@192-168-1-109:~/training_materials/developer/exercises/wordco _ □ ×
File Edit View Search Terminal Help

The filesystem under path '/user/training' is HEALTHY
[training@192-168-1-109 ~]$ hdfs dfs -ls
Found 2 items
drwxr-xr-x  - training supergroup          0 2020-08-03 13:30 shakespeare
drwxr-xr-x  - training supergroup          0 2020-08-03 13:24 wctest
[training@192-168-1-109 ~]$
[training@192-168-1-109 ~]$
[training@192-168-1-109 ~]$ hadoop classpath
/etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/./*:./usr/lib/hadoop-hdfs
/./usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/./*:./usr/lib/hadoop-yarn/li
b/*:/usr/lib/hadoop-yarn/./*:./usr/lib/hadoop-0.20-mapreduce/./usr/lib/hadoop-
0.20-mapreduce/lib/*:/usr/lib/hadoop-0.20-mapreduce/./:*
[training@192-168-1-109 ~]$ cd ~/training_materials/developer/exercises/wordcoun
t/src/
[training@192-168-1-109 src]$
[training@192-168-1-109 src]$ javac -classpath `hadoop classpath` stubs/*.java
[training@192-168-1-109 src]$ jar cvf wc.jar stubs/*.class
added manifest
adding: stubs/SumReducer.class(in = 1592) (out= 667)(deflated 58%)
adding: stubs/WordCount.class(in = 1543) (out= 837)(deflated 45%)
adding: stubs/WordMapper.class(in = 1705) (out= 711)(deflated 58%)
[training@192-168-1-109 src]$
```

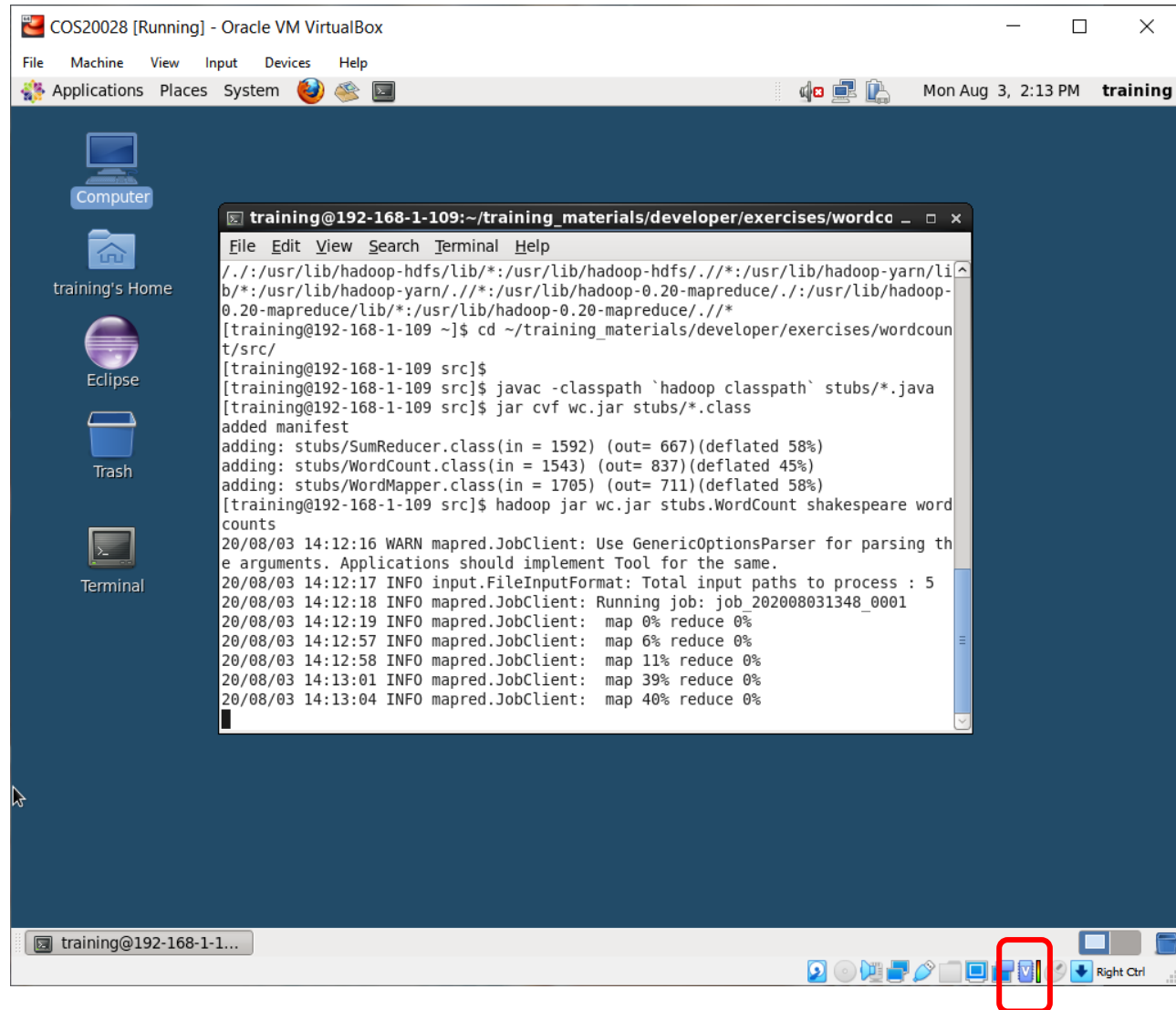
```

training@192-168-1-109:~/training_materials/developer/exercises/wordco _ □ ×
File Edit View Search Terminal Help
The filesystem under path '/user/training' is HEALTHY
[training@192-168-1-109 ~]$ hdfs dfs -ls
Found 2 items
drwxr-xr-x - training supergroup          0 2020-08-03 13:30 shakespeare
drwxr-xr-x - training supergroup          0 2020-08-03 13:24 wctest
[training@192-168-1-109 ~]$
[training@192-168-1-109 ~]$
[training@192-168-1-109 ~]$ hadoop classpath
/etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/./*/usr/lib/hadoop-hdfs
./:/usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/./*/usr/lib/hadoop-yarn/li
b/*:/usr/lib/hadoop-yarn/./*/usr/lib/hadoop-0.20-mapreduce/./:/usr/lib/hadoop-
0.20-mapreduce/lib/*:/usr/lib/hadoop-0.20-mapreduce/./*/
[training@192-168-1-109 ~]$ cd ~/training_materials/developer/exercises/wordcoun
t/src/
[training@192-168-1-109 src]$
[training@192-168-1-109 src]$ javac -classpath `hadoop classpath` stubs/*.java
[training@192-168-1-109 src]$ jar cvf wc.jar stubs/*.class
added manifest
adding: stubs/SumReducer.class(in = 1592) (out= 667)(deflated 58%)
adding: stubs/WordCount.class(in = 1543) (out= 837)(deflated 45%)
adding: stubs/WordMapper.class(in = 1705) (out= 711)(deflated 58%)
[training@192-168-1-109 src]$ hadoop jar wc.jar stubs.WordCount shakespeare word
counts

```

Execute MapReduce on Hadoop

- `hadoop jar wc.jar stubs.WordCount shakespeare wordcounts`



Wait Until Finished

- It will take a while to loop-over whole files.
- If you see the CPU bar is full, it is running.

```
training@192-168-1-109:~  
File Edit View Search Terminal Help  
20/08/03 14:14:15 INFO mapred.JobClient: Total time spent by all reduces waiting after reserving slots (ms)=0  
20/08/03 14:14:15 INFO mapred.JobClient: Map-Reduce Framework  
20/08/03 14:14:15 INFO mapred.JobClient: Map input records=175558  
20/08/03 14:14:15 INFO mapred.JobClient: Map output records=974078  
20/08/03 14:14:15 INFO mapred.JobClient: Map output bytes=8880434  
20/08/03 14:14:15 INFO mapred.JobClient: Input split bytes=594  
20/08/03 14:14:15 INFO mapred.JobClient: Combine input records=0  
20/08/03 14:14:15 INFO mapred.JobClient: Combine output records=0  
20/08/03 14:14:15 INFO mapred.JobClient: Reduce input groups=31809  
20/08/03 14:14:15 INFO mapred.JobClient: Reduce shuffle bytes=10828620  
20/08/03 14:14:15 INFO mapred.JobClient: Reduce input records=974078  
20/08/03 14:14:15 INFO mapred.JobClient: Reduce output records=31809  
20/08/03 14:14:15 INFO mapred.JobClient: Spilled Records=2862397  
20/08/03 14:14:15 INFO mapred.JobClient: CPU time spent (ms)=32520  
20/08/03 14:14:15 INFO mapred.JobClient: Physical memory (bytes) snapshot=1181302784  
20/08/03 14:14:15 INFO mapred.JobClient: Virtual memory (bytes) snapshot=4339527680  
20/08/03 14:14:15 INFO mapred.JobClient: Total committed heap usage (bytes)=859701248  
[training@192-168-1-109 src]$ cd ~  
[training@192-168-1-109 ~]$ hdfs dfs -get wordcounts
```

Harvest the Result

- `cd ~`
- `Hdfs dfs -get wordcounts`

```
training@192-168-1-109:~  
File Edit View Search Terminal Help  
[training@192-168-1-109 ~]$ less wordcounts/part-r-000000  
[training@192-168-1-109 ~]$
```

```
training@192-168-1-109:~  
File Edit View Search Terminal Help  
1      49  
10     1  
11th   1  
12th   1  
1s     1  
2      48  
2d     1  
2s     3  
3      29  
4      1  
4d     1  
5      1  
5s     1  
6      1  
6d     2  
6s     1  
7      1  
8      1  
8d     2  
9      1  
A      2027  
AARON  72  
ABATE  1  
wordcounts/part-r-000000
```

Check the Result

- `less wordcounts/part-r-000000`
- Use **PgUp** and **PgDn** to navigate in the file.
- Press **“q”** to exit.

- Since you only have one node in the cluster, you should only have 1 file starting with `part-`